

## Requirements Engineering & UML Use-Case Modelling

### PES1UG24CS144-Clinical Trial Subject Recruitment Engine

Actors: Trial Candidate, Clinical Researcher, Notification Service.

| Req ID  | Type                         | Description                                                                                                                                                                                               | Priority | Acceptance Criteria                                                                                                                                                                     | Rationale                                                                                |
|---------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| FR-001  | Functional                   | The system shall compare a trial candidate's questionnaire answers against the inclusion and exclusion criteria of the selected trial and mark the candidate as Eligible or Not Eligible.                 | High     | Pass: A candidate who meets an exclusion criterion is marked Not Eligible and the failed criteria are listed.<br>Fail: Such a candidate is marked Eligible.                             | Automated pre-screening is the main purpose of the system and prevents unsafe enrolment. |
| FR-002  | Functional                   | The system shall allow a Clinical Researcher to add and edit the inclusion and exclusion criteria of a trial.                                                                                             | High     | Pass: Criteria saved by the researcher are used in the very next screening.<br>Fail: Screening still uses the old criteria.                                                             | Trial criteria are revised during recruitment and must stay up to date.                  |
| FR-003  | Functional                   | The system shall display the informed consent form to an eligible candidate and store the candidate's electronic signature along with the form version, date and time.                                    | High     | Pass: The consent record contains the signature, form version and timestamp, and enrolment is blocked until it exists.<br>Fail: A candidate is enroled without a stored consent record. | Informed consent is legally required before a subject can join a trial.                  |
| FR-004  | Functional                   | The system shall allow a consented candidate to book study visits and allow the Clinical Researcher to mark each visit as Attended or Missed.                                                             | Medium   | Pass: A visit the subject did not attend appears as Missed in the visit compliance log.<br>Fail: The visit is shown as Attended or is not recorded at all.                              | Visit compliance has to be tracked to detect protocol deviations.                        |
| FR-005  | Functional                   | The system shall show the Clinical Resercher a report giving the number of candidates screened, found eligible, consented and enrolled for a selected trial.                                              | Medium   | Pass: Every count in the report matches the stored candidate records.<br>Fail: Any count differs from the stored records.                                                               | Researchers need to track recruitment progress against the enrolment target.             |
| NFR-001 | Non-functional (Security)    | The system shall record an audit trail entry containing the user, date, time, old value and new value for every change made to a candidate, eligibility or consent record, as required by 21 CFR Part 11. | High     | Pass: Every record change creates an audit entry, and stored entries cannot be edited or deleted.<br>Fail: Any record change leaves no audit entry.                                     | Regulators require proof that clinical records have not been altered.                    |
| NFR-002 | Non-functional (Performance) | The system shall display the eligibility result within 3 seconds for at least 95% of submissions while 200 candidates are using the system at the same time.                                              | High     | Pass: A load test with 200 concurent users returns 95% of results in 3 seconds or less.<br>Fail: More than 5% of results take longer than 3 seconds.                                    | Slow screening feedback causes candidates to abandon the application.                    |